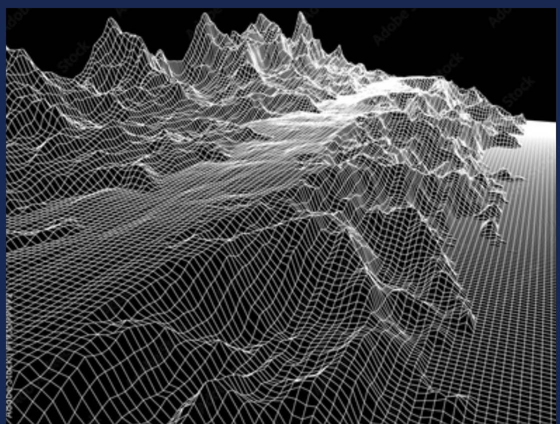
FRAMEWORK OVERVIEW

A custom-built C++17 and OpenGL 3.3 graphics and AI engineering testbed designed to experiment with rendering architecture, high-density scene generation, real-time agent navigation, and persistent networking.

Core Technical Highlights:

- Custom Deferred G-Buffer Shading Pipeline
- 1M+ Grass Blade GPU Instancing & Culling
- Grid-Based A* Pathfinding Engine
- Finite State Machine (FSM) Agent Behaviors
- Winsock2 TCP & MongoDB Asynchronous State Storage



MORE ON THIS PROJECT

PROJECT REPOSITORY & DEMO

Scan to View Source Code & Demo

NOTE:THIS ONLY CONTAIN THE BASE VERSION

SUPERVISOR

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ENGINEERING HIGHLIGHTS

- Target Workload: 1M+ GPU-instanced grass blades @ real-time frame rates
- AI Pathfinding: Zero-allocation runtime grid A* navigation
- Persistence: Non-blocking TCP socket state sync via MongoDB

TECHNOLOGICAL UNIVERSITY (MANDALAY)

DEPARTMENT OF COMPUTER ENGINEERING AND INFORMATION TECHNOLOGY

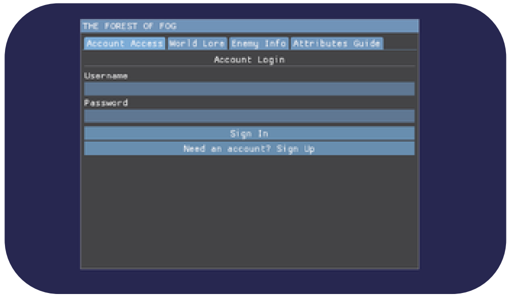
MULTI-USER 3D SIMULATION ENVIRONMENT WITH EFFICIENT PATH-FINDING ALGORITHM

GRAPHICS & RENDERING

Custom C++17/OpenGL deferred shading pipeline built to maximize frame rates in high-density 3D environments.

Key Specs:

- Multi-Attachment G-Buffer
- Dynamic Point Lighting
- GPU Hardware Skinning
- 1M+ Grass/Tree Instancing



LOG IN/SIGN UP UI

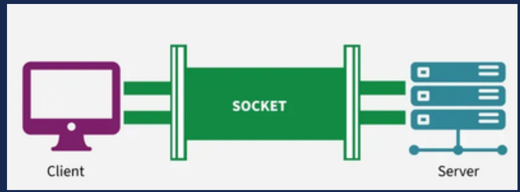
THE GAME VISUAL

NETWORK & DATABASE

Asynchronous Winsock2 TCP architecture paired with MongoDB for non-blocking state persistence.

Key Specs:

- Socket Networking
- Dynamic Persistence
- Stutter-Free Execution

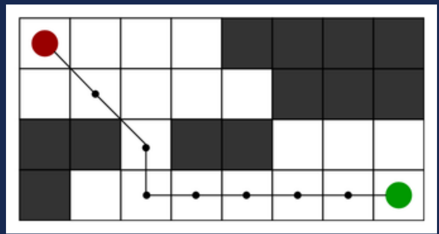


A* PATHFINDING

Static-memory grid navigation for real-time obstacle avoidance and agent tracking.

Key Specs:

- Grid Routing
- Zero Runtime Allocation
- Dynamic Target Tracking

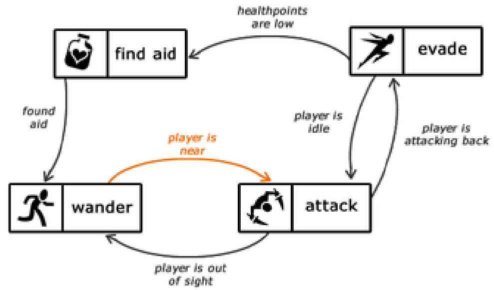


FINITE STATE MACHINE (FSM)

Dynamic decision engine controlling combat transitions and agent state execution.

Key Specs:

- Idle/Run/Attack States
- Ability Cooldowns
- Integrated Pathing Logic



TEAM MEMBERS

- MG AUNG MYO PAI
- MA KHIN YADANAR WIN
- MG HLWAN MOE AUNG
- MA THOON THIRI SWE
- MG NAY PHONE MYINT
- MG KAUNG KHANT KO KO